

PROJECT

6.2

AURORA DATABASE WITH EC2

DATABASE SERIES

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Objective

Learn to create and connect an Amazon Aurora MySQL database with an EC2 instance for secure data access. Gain practical experience in managing cloud resources within a private network setup.

This project helps you understand how Aurora scales efficiently and how EC2 can serve as a client for secure database interactions within a VPC.

Services

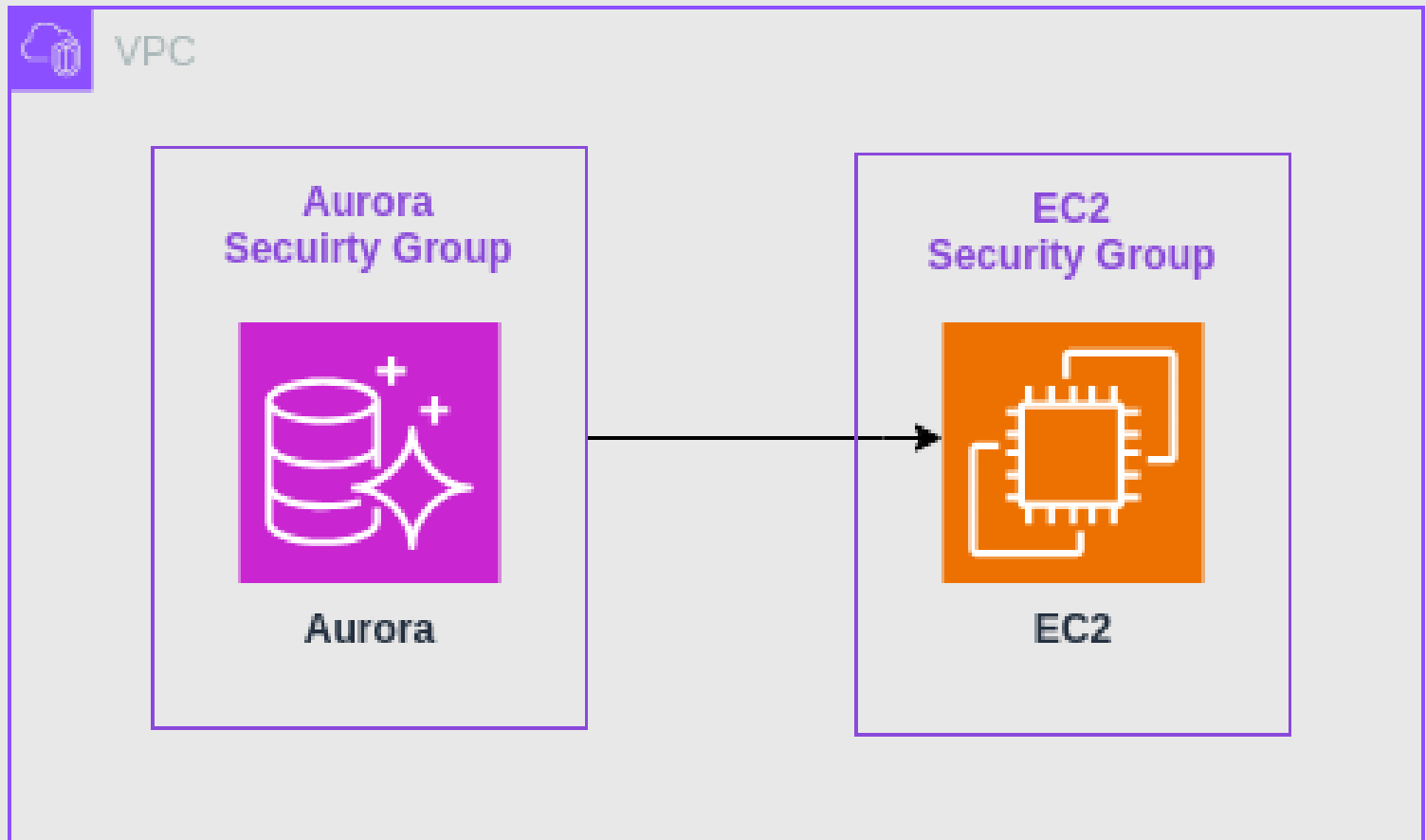
Amazon Aurora (MySQL-Compatible) - A high performance relational database service used to store and manage structured data.

Amazon EC2 - A virtual server used to access and interact with the Aurora database securely.

IAM (Identity and Access Management) - Used to manage user access and permissions for secure operations.

Amazon VPC - Creates a private and isolated cloud network where EC2 and Aurora can securely communicate using internal IPs.

Architecture Diagram



Aurora and EC2 are both launched within the same VPC to ensure private and secure communication. Each is placed in a subnet and protected using security groups that allow controlled access between them. The EC2 instance acts as a client machine to connect to the Aurora database using internal networking, without public exposure.

Steps to Implement 1

1. Login with IAM User:

Start by logging into your AWS account using a secure IAM user. Make sure the user has permissions for RDS, EC2, and VPC. This helps you follow best practices and avoid using the root account.

2. Create an Aurora MySQL Database:

Go to the RDS console and launch a new Amazon Aurora MySQL database. Choose the instance size, storage settings, and VPC/subnet group. Keep the database in a private subnet and take note of the endpoint.

3. Launch an EC2 Instance:

Navigate to the EC2 dashboard and create a new instance within the same VPC. Choose an Amazon Linux or Ubuntu AMI for easier setup. Add a key pair and make sure the security group allows outbound access to RDS.

4. Finish Creating Aurora Database:

Ensure the Aurora cluster is available and running. Modify the Aurora security group to allow inbound MySQL traffic (port 3306) from the EC2 instance's security group. This opens a secure path for communication.

Steps to Implement 2

5. Connect from EC2 to Aurora:

SSH into your EC2 instance using the key pair from step 3. Install MySQL client if needed and connect to the Aurora DB using its endpoint, port, and credentials. Test the connection to ensure everything works.

6. Manage and Query the Database:

Once connected, you can run SQL queries, create tables, or insert sample data from the EC2 terminal. This simulates real-world app-to-database interaction. Keep your database private and protected at all times.

7. Monitor and Secure:

Use CloudWatch to monitor database performance and EC2 metrics. Regularly review IAM roles, VPC flow logs, and security groups. Make sure unused ports are closed to keep your environment safe.

8. Clean Up Resources:

After testing, terminate your EC2 instance and delete the Aurora database to avoid charges. Also remove any temporary IAM roles or key pairs. Always clean up to manage your cloud costs efficiently.

Story Time

VPC (Your Cloud Fortress):

Imagine building a secret digital fortress that's your VPC! Inside these guarded walls, only trusted services like EC2 and Aurora are allowed to live and talk. It's like a private castle where every door is guarded by a loyal knight (security group).

Aurora (Your Smart Library):

Next, you set up Aurora your cloud city's smartest and fastest library! It holds all the knowledge (data) and responds to questions super fast. But remember, it won't talk to strangers only trusted friends with special access can open its doors.

EC2 (Your Data Explorer):

Now bring in EC2, your adventurous data explorer, who lives inside the same fortress. EC2 is clever and quick, perfect for talking to Aurora and getting information. But only if their security badges (groups) match!

IAM (Your Royal Gatekeeper):

Every kingdom needs a gatekeeper, and IAM is yours! It gives keys (permissions) to only those who deserve it. Without IAM, no one not even your explorer can move safely or manage your fortress tools.

Security Groups (Castle Guards):

Finally, the castle guards (security groups) control who EC2 and Aurora can speak to. If the rules are set right, they become best friends and chat all day. But set them wrong, and it's total silence!

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THANK YOU!

